



INSPIRE FORENSICS

(Recognised by the Indian Society of Forensic Science)

CUET-UG 2026

SAMPLE PAPER

A National Level Socio-Forensic Voluntary Student Led Educational Firm

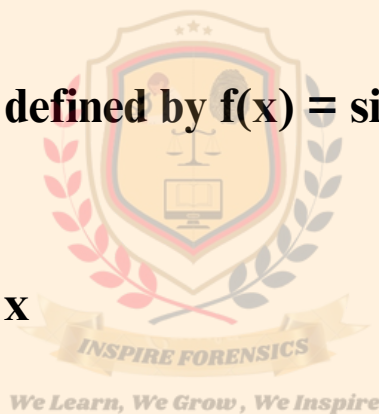
Core Mathematics / Applied Mathematics (319)

1. Consider the relation R defined on the set $A = \{1, 2, 3\}$ as $R = \{(1,1), (2,2), (3,3), (1,2), (2,3), (1,3)\}$. Which of the following properties does R exhibit?

- A. Reflexive only
- B. Reflexive and transitive
- C. Symmetric and transitive
- D. Neither symmetric nor transitive

2. If a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \sin x + x$, what is $f(f(x))$?

- A. $2 \sin x + x$
- B. $\sin^2 x + x^2$
- C. $\sin(\sin x + x) + \sin x + x$
- D. $\sin^2 x + 2 \sin x + x$



3. What is the value of the expression: $\sin(\tan^{-1}x) - \cot^{-1}x \sin(\sin^{-1}x + \cos^{-1}x)$, for $x \in [-1, 1]$?

- A. 1.0
- B. $\pi/2$
- C. 3.1
- D. $-\pi/2$

4. Match List I with List II and choose the correct answer:

LIST I

- A. $\sin^{-1}x + \cos^{-1}x, x \in [-1, 1]$**
- B. $\tan^{-1}3 - \cot^{-1}(-3)$**
- C. $\cos^{-1}(\cos 13\pi/6)$**
- D. $\sin^{-1}(\sin 12)$**

LIST II

- I. $\pi/2$**
- II. $\pi/6$**
- III. $\pi/2$**
- IV. 12**

- A. A-I, B-II, C-III, D-IV**
- B. A-II, B-I, C-III, D-IV**
- C. A-I, B-III, C-II, D-IV**
- D. A-IV, B-II, C-III, D-I**



5. The function $f(x) = x - 1/x(x^2 - 1)$, $x \neq 1$, $f(1) = 1$, is discontinuous at:

- A. Exactly one point**
- B. Exactly two points**
- C. Exactly three points**
- D. No point**

6. What is the derivative of $\sin(\tan^{-1}e^{2x})$ with respect to x ?

- A. $2e^{2x} \sin(\tan^{-1}e^{2x}) / (1 + e^{4x})$**
- B. $2e^{2x} \cos(\tan^{-1}e^{2x}) / (1 + e^{4x})$**
- C. $2e^{2x} \sin(\tan^{-1}e^{2x}) / (1 + e^{2x})$**
- D. $2e^x \cos(\tan^{-1}e^{2x}) / (1 + e^x)$**

7. If $1-x^2+1-y^2=a(x-y)$, then $\frac{dy}{dx}$ is:

- A. $1-x^{21}-y^2$
- B. $1-y^{21}-x^2$
- C. $1-x^{21}-y^2$
- D. $1+x^{21}-y^2$

8. A manufacturer can sell x items at a price of ₹ $3x-5$ each. The cost price of x items is ₹ $2x^2+5x$. If the manufacturer has to sell items to get no profit and no loss, then the number of items x is:

- A. $x=10$
- B. $x=30$
- C. $x=0$
- D. $x=-10$

9. The approximate volume of a cube of side a meters on increasing the side by 4% is:

- A. $1.04a^3m^3$
- B. $1.004a^3m^3$
- C. $1.12a^3m^3$
- D. $1.112a^3m^3$



10. The maximum slope of the curve $y=-x^3+3x^2+9x-27$ is:

- A. 1.0
- B. 12
- C. 16
- D. 32

11. Match the integrals in LIST I with their results in LIST II and choose the correct answer:

LIST I

- A. $\int \sin x \frac{1}{1+\cos x} dx$
- B. $\int \tan x \frac{1}{1-\tan x} dx$
- C. $\int \tan^{-1} x dx$
- D. $\int \frac{1}{x} + \log x dx$

LIST II

- I. $\tan^{-1} x + c$
- II. $\log(\log x + 1) + C$
- III. $-\log|1+\cos x| + C$
- IV. $x^2 \log|\cos x - \sin x| + C$

- A. A-III, B-IV, C-I, D-II
- B. A-II, B-I, C-III, D-IV
- C. A-I, B-II, C-III, D-IV
- D. A-IV, B-I, C-II, D-III



12. Evaluate the integral: $\int (1+x+x^{21}+x^2 \tan^{-1} x) dx$.

- A. $x + \tan^{-1} x + c$
- B. $\tan^{-1} x - x + c$
- C. $\tan^{-1} x + c$
- D. $x \tan^{-1} x + c$

13. The area of the region bounded by the parabola $y^2=4ax$ and its latus rectum is:

- A. $4a^{\frac{2}{3}}$ sq. units
- B. $8a^{\frac{2}{3}}$ sq. units
- C. $2a^{\frac{2}{3}}$ sq. units
- D. $9a^{\frac{2}{5}}$ sq. units

14. The area of the region bounded by the lines $x=2y+3$, $x=0$, $y=0$ and $y=-1$ is:

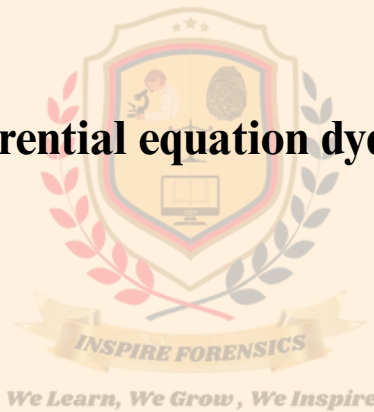
- A. 4 sq. units
- B. 6 sq. units
- C. 8 sq. units
- D. 32 sq. units

15. The particular solution of the differential equation $\log(dy/dx)=x+y$, given that when $x=0$, $y=0$, is:

- A. $e^x+y=2$
- B. $e^x+ey=2$
- C. $e^x+ey=2$
- D. $e^x-ey=2$

16. The solution of the differential equation $dy/dx=(1+x^2)(1+y^2)$ is:

- A. $\tan^{-1}y=x^{33}+x+c$
- B. $\tan^{-1}y=x-x^{33}+c$
- C. $\tan^{-1}y=x^2+x^{33}+c$
- D. $\tan^{-1}y=x^2-x^{33}+c$



17. If a line makes angles 90° , 60° and θ with x , y and z axis respectively, where θ is acute, then the value of θ is:

- A. $\pi/6$
- B. $\pi/4$
- C. $\pi/3$
- D. $\pi/2$

18. Match the items in LIST I with those in LIST II and choose the correct answer:

LIST I

- A. The area of parallelogram determined by vectors $2\vec{i}$ and $3\vec{j}$.
- B. The value of $(\vec{i} \times \vec{j}) \cdot \vec{k} + (\vec{j} \times \vec{k}) \cdot \vec{i}$.
- C. The value of λ for which the vectors $2\vec{i} - 3\vec{j} + 4\vec{k}$ and $\vec{a} - 6\vec{j} + 8\vec{k}$ are collinear.
- D. The value of λ for which the vectors $2\vec{i} + \lambda\vec{j} + \vec{k}$ and $2\vec{i} - 4\vec{j} + 2\vec{k}$ are perpendicular.

LIST II

- I. 2
- II. 4
- III. 0
- IV. 6

- A. A-I, B-II, C-III, D-IV
- B. A-II, B-I, C-III, D-IV
- C. A-III, B-IV, C-II, D-I
- D. A-IV, B-I, C-II, D-III



19. Consider the following statements regarding equations of lines and planes:

- A. Equation of the line passing through the point (1, 2, 3) and parallel to the vector $3\vec{i} + 2\vec{j} - 2\vec{k}$ is $x - 13 = y - 22 = z - 3 - 2$.
- B. Equation of line passing through (1, 2, 3) and parallel to the line given by $x + 43 = y - 56 = z - 12$ is $x - 13 = y - 26 = z - 32$.
- C. Equation of plane passing through the origin and (5, -2, 3) is $x^5 = y - 2 = z^3$.
- D. Equation of plane with intercepts 2, 3 and 4 on x, y and z-axis respectively is $2x + 3y + 4z = 1$.

Which of the following statements are correct?

- A. A, C only
- B. A, D only
- C. C, D only
- D. A only

20. The angle between the line $x+23=y-32=z+56$ and the plane $2x+10y-11z=5$ is:

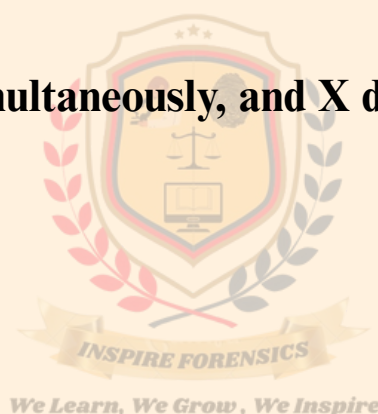
- A. $\cos^{-1}(821)$
- B. $\sin^{-1}(821)$
- C. $\cos^{-1}(2182)$
- D. $\sin^{-1}(2182)$

21. The feasible region of an LPP is bounded in the first quadrant. Which of the following statements about its solution is correct?

- A. It is bounded but has no solution.
- B. It is unbounded but has a solution.
- C. It is bounded and has a solution.
- D. It is unbounded and has no solution.

22. If two dice are thrown simultaneously, and X denotes the number of sixes, then the variance of X is:

- A. 518
- B. 718
- C. 13
- D. 23



23. Probabilities to solve a specific problem by A, B and C are 12, 13 and 14 respectively. The probability that at least one will solve the problem is:

- A. 124
- B. 14
- C. 2324
- D. 34

24. Which of the following statements about matrices is NOT correct?

- A. A row matrix has only one row.
- B. A diagonal matrix has all diagonal elements equal to zero.
- C. A symmetric matrix is a square matrix satisfying certain conditions.
- D. A skew-symmetric matrix has all diagonal elements equal to zero.

25. If the matrix $A = [\cos\theta \sin\theta -\sin\theta \cos\theta]$, then A^2 is equal to:

- A. $[\cos 2\theta \sin 2\theta -\sin 2\theta \cos 2\theta]$
- B. $[\cos 2\theta \sin 2\theta -\sin 2\theta \cos 2\theta]$
- C. $[\cos 2\theta \sin\theta -\sin\theta \cos 2\theta]$
- D. $[\cos\theta +\sin\theta \cos\theta -\sin\theta \sin\theta -\cos\theta \cos\theta +\sin\theta]$

26. If the matrix $A = [0x+y \ 13z \ 2x-y-20]$ is skew-symmetric, then the values of x, y, and z are:

- A. $x=2, y=1, z=0$
- B. $x=2, y=-3, z=8$
- C. $x=-2, y=-1, z=0$
- D. $x=-2, y=3, z=-1$

27. If A is a square matrix of order 3, then $|\text{adj } A|$ is equal to:

- A. $|A|$
- B. $|A|^2$
- C. $|A|^3$
- D. $3|A|$



28. If three points $A(a^1, b^1)$, $B(a^2, b^2)$ and $C(a^3, b^3)$ are collinear and $D = |a_1 b_1 1 a_2 b_2 1 a_3 b_3 1|$, then:

- A. $D = 0$
- B. $D = \pm 1$
- C. $D^2 = 0$ or 1
- D. $D = (a^1 + a^2 + a^3) - (b^1 + b^2 + b^3)$

29. If $A = [\cos\theta \sin\theta 0 -\sin\theta \cos\theta 0 0 0 1]$ and B is a square matrix of order 3, then $|AB|$ is equal to:

- A. $|B|^2$
- B. $|B|$
- C. $\sin 2\theta |B|$
- D. $\cos 2\theta |B|$

30. The range of the function $f(x)=13-\sin 4x$ is:

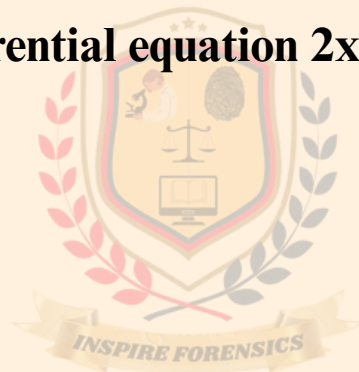
- A. [14,12]
- B. [12,1]
- C. [14,34]
- D. [24,13]

31. The equation of the tangent to the curve $y=x^2-2x-3$ which is perpendicular to the line $x+2y+3=0$, is:

- A. $4x-2y=7$
- B. $2x-y=7$
- C. $3x-y=5$
- D. $4x-2y=5$

32. The solution of the differential equation $2x dy/dx + y = 14x^3$, $x > 0$, is:

- A. $y=2x^3+c_1x$
- B. $y=x^3+c_1x^2$
- C. $y=2x^3+c_1x^2$
- D. $y=x^3+c_1x$



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33. If the vectors $\vec{a}=\hat{i}-3\hat{j}+2\hat{k}$, $\vec{b}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=3\hat{i}+5\hat{j}-2\lambda\hat{k}$ are coplanar, then λ is equal to:

- A. -1
- B. 2.1
- C. -2
- D. 4.2

34. A coin is tossed 7 times. The probability of getting at least 4 heads is:

- A. 58
- B. 34
- C. 14
- D. 12

35. If $57 \equiv x \pmod{5}$, then the least positive value of x is:

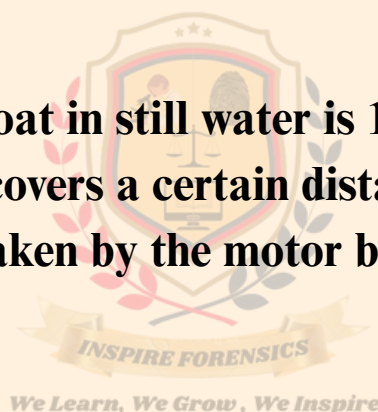
- A. 1.57
- B. 2.5
- C. 3.4
- D. 4.2

36. Pure honey costs ₹300 per litre. A shopkeeper adds water to 10 litres of pure honey and sells the resulting syrup at ₹250 per litre. The quantity of water added by the shopkeeper is:

- A. 1.2 litres
- B. 2.5 litres
- C. 3.3 litres
- D. 4.1.5 litres

37. The speed of a motor boat in still water is 14.4 times the speed of the current. If the motor boat covers a certain distance upstream in 6 hours 25 minutes, then the time taken by the motor boat to come back is:

- A. 1. 5 hours 35 minutes
- B. 2. 5 hours 25 minutes
- C. 3. 5 hours 10 minutes
- D. 4. 5 hours 55 minutes



38. The longest side of a triangle is four times the shortest side. The third side of the triangle is 3cm shorter than the longest side. If the perimeter of the triangle is at least 69 cm, then its:

- A. Shortest-side < 8 cm
- B. Shortest-side > 8 cm
- C. Shortest-side ≤ 8 cm
- D. Shortest-side ≥ 8 cm

39. Three persons A, B and C enter into a partnership to run a business. They invested their capitals in the ratio 4:5:6. After 5 months, B increases his share by 40%. If the total profit at the end of a year is ₹50,550, then A's share in the profit is:

- A. ₹8,000**
- B. ₹10,000**
- C. ₹20,000**
- D. ₹12,000**

40. Match List I with List II and choose the correct answer:

LIST I

- A. The set of values of decision variables which do not satisfy all the constraints and non-negativity condition of an LPP is called**
- B. In an LPP, the objective function is always**
- C. In an LPP, the linear inequalities on variables are called**
- D. The feasible region for an LPP is always**

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LIST II

- I. Linear**
- II. Convex polygon**
- III. Infeasible solution**
- IV. Constraints**

- A. A-III, B-I, C-IV, D-II**
- B. A-IV, B-III, C-I, D-II**
- C. A-III, B-I, C-II, D-IV**
- D. A-I, B-II, C-IV, D-II**

41. Let $f(x + y) = f(x) f(y)$ for all x and y . Suppose that $f(3) = 3$ and $f'(0) = 11$, then $f'(3)$ is given by:

- A. 28
- B. 22
- C. None of these
- D. 33

42. The domain of $\sin^{-1}(2x)$ is:

- A. $[1, 1]$
- B. $[-1, 1]$
- C. $[-1/2, 1/2]$
- D. $[-2, 2]$

43. A and B are two non-zero square matrices such that $AB = O$ (the null matrix). Then:

- A. Both A and B are singular
- B. Either A or B is singular
- C. Neither matrix is singular
- D. None of these



44. If K is a scalar and A is an $n \times n$ square matrix, then the determinant $|KA|$ is equal to:

- A. $K * |A|^n$
- B. $K * |A|$
- C. $K^n * |A^n|$
- D. $K^n * |A|$

45. Let $f(x) = |x| + |x - 1|$, then:

- A. $f(x)$ is continuous at $x = 0$, as well as at $x = 1$
- B. $f(x)$ is continuous at $x = 0$, but not at $x = 1$
- C. $f(x)$ is continuous at $x = 1$, but not at $x = 0$
- D. None of these

46. The value of the Limit as x approaches e of $[(\log x - 1) / (x - e)]$ is:

- A. e
- B. 1
- C. e^{-1}
- D. None of these

47. Find dy/dx of $y = e^{(x + e^{(x + e^{(x + \dots \text{infinity})}))})}$:

- A. $x + y$
- B. xy
- C. $y / (1 - y)$
- D. $1 + x + y$

48. If $f(x) = \tan x + e^{-2x} - 7x^3$, then what is the value of $f'(0)$?

- A. -2
- B. -1
- C. 0
- D. 3



49. The order and degree of the differential equation $(d^3y/dx^3)^{2/3} + 4 - 3(d^2y/dx^2) + 5(dy/dx) = 0$ are respectively:

- A. 1, 2
- B. 3, 2
- C. 2, 2
- D. 1, 3

50. The value of the Integral from 0 to 8 of $|x - 5| dx$ is:

- A. 17
- B. 15
- C. 21
- D. 12

51. The area of the region enclosed by $y = x^2$ and $2y = x$ is:

- A. $1/48$
- B. $1/24$
- C. $1/16$
- D. $1/12$

52. What is the Integral from 0 to $\pi/2$ of $[(\sin^3 x) / (\sin^3 x + \cos^3 x)] dx$?

- A. $\pi/4$
- B. $\pi/2$
- C. π
- D. 0

53. A can hit a target 4 times in 5 shots, B hits 3 times in 4 shots, and C hits twice in 3 shots. They fire together. The probability that at least two shots hit the target is:

- A. $5/6$
- B. $(1/3)^2$
- C. $13/30$
- D. None of these



54. Find the direction ratios of the line determined by the planes $2x - y - z = 2$ and $x + 2y - 3z = 11$

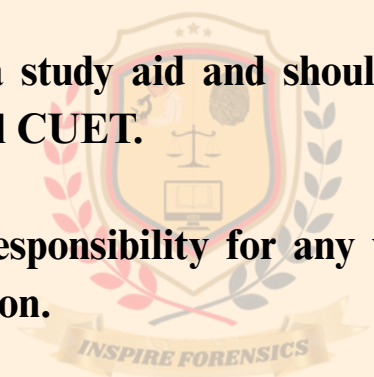
- A. 1, 2, 2
- B. 1, 1, 1
- C. 2, 2, 3
- D. 2, 1, 1

55. If the points with position vectors $60\mathbf{i} + 3\mathbf{j}$, $40\mathbf{i} - 8\mathbf{j}$, and $a\mathbf{i} - 52\mathbf{j}$ are collinear, then the value of a is:

- A. -40
- B. 20
- C. 40
- D. -20

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Answer Key

Question number	Answer Key		Question number	Answer Key
1	B		19	D
2	C		20	B
3	A		21	C
4	A		22	A
5	B		23	C
6	A		24	B
7	C		25	A
8	A		26	C
9	A		27	B
10	C		28	A
11	A		29	B
12	A		30	A
13	B		31	A
14	A		32	D
15	C		33	C
16	A		34	B
17	C		35	B
18	B		36	D

Answer Key

Question number	Answer Key
37	A
38	C
39	B
40	A
41	D
42	C
43	B
44	D
45	A
46	C
47	C
48	B
49	B
50	A
51	A
52	A
53	A
54	B
55	A